

9 (a) On a single Argand diagram, sketch the loci defined by

- $\arg(z - 2) = \frac{3}{4}\pi,$

- $|z| = |z + 2 - i|.$

[4]

(b) In this question you must show detailed reasoning.

The point of intersection of the two loci in part (a) represents the complex number w .

Find w , giving your answer in exact form.

[5]